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New palaeomagnetic and rock magnetic results on Mesoproterozoic kimberlites from the Eastern Dharwar craton, southern India: Towards constraining India's position in Rodinia

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ABSTRACT

New palaeomagnetic and rock magnetic results are reported for 1.1 Ga Mesoproterozoic kimberlites from the distinct fields of Wajrakarur (WKF), Narayanpet (NKF) and Raichur (RKF) occurring within the Eastern Dharwar craton, southern India. In all, 35 oriented block samples were collected from ten kimberlite pipes and our results constitute first of their kind for the kimberlites from NKF and RKF. Palaeomagnetic data reveal that characteristic remanent magnetization (ChRM) directions of the studied kimberlites show a mean declination of 39 and mean inclination of −16 ($N = 4$, $k = 37.6$ and $\alpha_{95} = 15.2$) and yield a virtual geomagnetic pole at 44.5°N, 195.4°E ($d_p = 8.0$, $d_m = 15.6$). This new pole position is identical to the earlier reported palaeomagnetic pole that lies at 36.8°N, 212.5°E for some Wajrakarur kimberlites and other such global results at 1100 Ma from Umkondo dolerites, Kalahari craton, southern Africa. The newly obtained palaeomagnetic pole also correlates well with the virtual geomagnetic pole (VGP) generated from the 1.1 Ga Majhgawan kimberlite, Mahoba dyke and the Bhander–Rewa sedimentary Group in the Upper Vindhyan sequence within the Bundelkhand craton of Northern India. Our results support a Mesoproterozoic closure age of the upper sedimentary horizons for the 'Purana' sedimentary basins and provide evidence for accretion of the northern and southern Indian cratonic blocks prior to 1.1 Ga. Our study also demonstrates that in ~1.1 Ga palaeomagnetically based Rodinia reconstructions, India occupies a lower palaeolatitudinal position, was much separated from Australia and that East Gondwana very likely did not form an assembly until the terminal Neoproterozoic.

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1. Introduction

Continental intra-plate alkaline magmas such as kimberlites are rare and exotic rocks that are known to be generated at great depths and serve as "time-capsules' in a tectonic frame-work by revealing periods of supercontinent assembly, their break-ups as well as periods of continental stability (Jelsma et al., 2009). World-wide kimberlites are now known to be virtually confined to the ancient shield areas or mobile belts that are underlain by thick cratonic keels or lithospheric roots (e.g., Clifford, 1966; Jaques et al., 1990; Viljoen, 2002). There are at least four major periods during which the kimberlite emplacement has occurred i.e., the Mesoproterozoic, the Palaeozoic, the Mesozoic and the Cenozoic (e.g., Allsopp et al., 1989; Heaman et al., 2004; Jelsma et al., 2009). A variety of models involving mantle plumes (e.g., Torsvik et al., 2010; Collerson et al., 2010; Chalapathi Rao et al., 2011; Chalapathi Rao and Lehmann, 2011), extension (e.g., Vearncombe and Vearncombe, 2002; Key et al., 2007) and even subduction (e.g., McCandless, 1999; Chalapathi Rao et al., 2010a) have been invoked for the eruption of kimberlites.

Kimberlites (some of them diamondiferous) and lamproites are widespread in five areas within the Eastern Dharwar craton (EDC) as well as within and around the Palaeo-Mesoproterozoic Cuddapah basin (Fig. 1) in southern India and are confined to distinct fields viz., Wajrakarur kimberlite field (WKF), Narayanpet kimberlite field (NKF), Raichur kimberlite field (RKF), Krishna lamproite field (KLF) and Cuddapah lamproite field (e.g., Paul et al., 2006; Kumar et al., 2007; Chalapathi Rao, 2008; Chalapathi Rao and Srivastava, 2009; Chalapathi Rao et al., 2010b). Likewise, a number of highly diamondiferous ultramafic rocks having affinities towards lamproites are known from Majhgawan, Hinota and Bunder areas in the Aravalli craton, Central India (e.g., Chalapathi Rao, 2005; Gregory et al., 2006; Masun et al., 2009; Meert et al., 2010). Some of these kimberlites from the Eastern Dharwar craton and lamproites

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from Central India are known to be associated with the formation of Rodinia supercontinent (Kumar et al., 2007; Meert et al., 2010), that existed somewhere between 1000 and 750 Ma ago (Dalziel, 1991). The Rodinia supercontinent also included many of the present day continents from southern hemisphere including Australia and Antarctica that were reorganized to form Gondwanaland subsequently. India's position in the Rodinia supercontinent configuration; however, constitutes a matter of debate. According to one model, India was juxtaposed with Australia and East Antarctica in a Rodinian fit (e.g., Hoffman, 1991; Torsvik et al., 1996; Weil et al., 1998; Pradhan et al., 2010) whereas in the other models it has been altogether excluded (e.g., Torsvik et al., 2001; Meert and Torsvik, 2003). Therefore, palaeomagnetism of Mesoproterozoic kimberlites in the Eastern Dharwar craton hold much promise in constraining India's position in the Rodinia configuration and contributes towards a better understanding of the late Mesoproterozoic supercontinent tectonics. A refined palaeomagnetic pole of the EDC kimberlites is also important in constraining the proximity of the northern and southern blocks of India as their emplacement age of 1100 Ma (Kumar et al., 2007; Chalapathi Rao et al., in press) is strikingly similar to that of the Majhgawan diamondiferous pipe of Bundelkhand craton, northern India (Gregory et al., 2006).

In this paper we report a new pole position for kimberlites from the WKF, NKF and RKF fields of Eastern Dharwar craton and explore its significance in contributing to the on-going debate on the closure age of 'Purana' sedimentary basins in peninsular India as well as in the context of Rodinia tectonics.

2. Geological setting and geochronology of kimberlites from the Eastern Dharwar craton

The Dharwar craton is made of two separate cratons, an east and west, with major differences in lithology and ages. The western boundary of the Eastern Dharwar Craton (EDC) is poorly defined and is constrained to a 200 km wide lithologic transitional zone from the peninsular gneisses of the Western Dharwar craton to the Closepet Granite. The EDC is bounded to the north by the Deccan traps and the Bastar craton, to the east by the Eastern Ghats mobile belt, and to the south by the Southern Granulite terrene (Balakrishnan et al., 1999; Meert et al., 2010). The EDC is composed of the Dharwar batholith (dominantly granitic), greenstone belts, intrusive volcanics, and Mesoproterozoic to more recent sedimentary basins (Naqvi and Rogers, 1987; Ramakrishnan and Vaidyanadhan, 2008; Meert et al., 2010).

Some of the WKF pipes viz. Lattavaram, Wajrakarur and Muligiripalle have earlier been studied palaeomagnetically by Miller and Hargraves (1994). The Wajrakarur samples were unstable and yielded no useful data whereas the Muligiripalle and Lattavaram pipes yielded similar palaeomagnetic data. However, a careful examination of their plots show that more than 95% of the remanence was random and the 'characteristic' direction was obtained by extrapolation of what they envisaged as stable endpoints. The α_{95}'s were quite large (>20°) and therefore, these poles should be viewed with caution. There are excellent radiometric age constraints on kimberlite emplacement in many of these fields. The WKF is the best dated with Rb–Sr ages (phlogopite) forming a tight cluster between 1091 and 1102 Ma. These dates are additionally supported by a U–Pb perovskite age of 1079 Ma for the Muligiripalle pipe (Miller and Hargraves, 1994) but the analytical details were not provided by these authors. However, a recent U–Pb (perovskite) age of 1124 ± 5 Ma has been reported by Kumar et al. (2007). Together with several U–Pb (perovskite) ages of 1079 ± 11 (Pipe-5; Muligiripalle), 1125 ± 16 (Pipe-7, Venkatampalle), 1117 ± 14 (Pipe-13, Tummatpalle), 1120 ± 18 (CC-4; Gollapalle), 1128 ± 12 (CC5/1; Gollapalle) and 1101 ± 11 (KL-3; Kalyandurg) recently obtained by

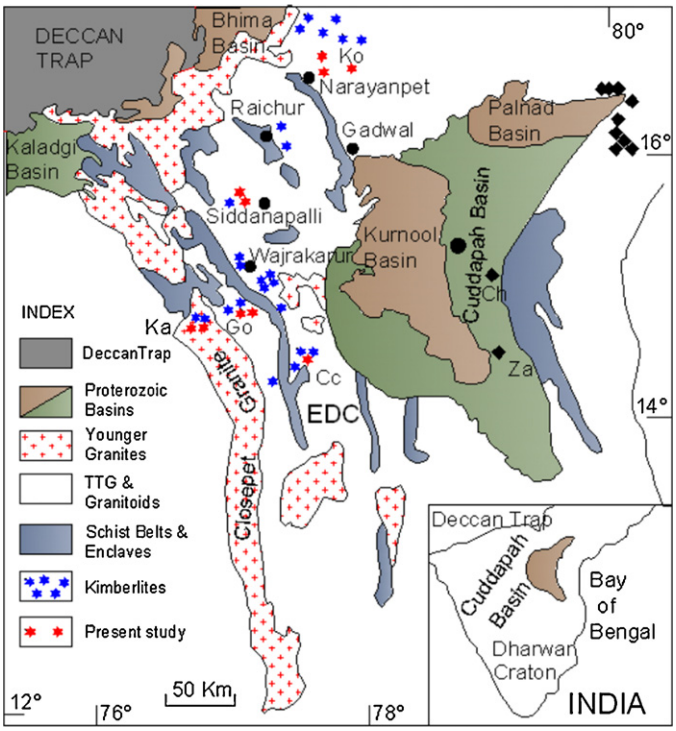


Fig. 1. Geological map of south India showing the locations of kimberlites and lamproites. (adapted from Chalapathi Rao et al., 2010c and references therein). Symbols: red stars= kimberlites of this study; blue stars = other kimberlites; diamonds = lamproites. Abbreviations: EDC, Eastern Dharwar craton; Ko, Kotakonda; Ka, Kalyandurg; Cc, Chigicherla; Go, Gollapalli; Za, Zangamrajupalli; Ch, Chelima. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of the article.)

us corroborate a widespread tectono-magmatic kimberlite event at 1100 Ma in the Eastern Dharwar craton (Chalapathi Rao et al., in press).

The newly discovered cluster at Siddanpalli (RKF) yielded an Rb–Sr whole rock mineral isochron age of 1093 ± 4 Ma (Kumar et al., 2007) that is also indistinguishable from the 1093 ± 18 U–Pb (perovskite) age (Chalapathi Rao et al., in press). Likewise, Rb–Sr ages on kimberlites from the Kotakonda and Mudabid kimberlite intrusions (NKF) yielded coherent ages of 1084 ± 14 and 1098 ± 12 Ma (Kumar et al., 2001) that are also similar to the recently obtained U–Pb (perovskite) ages of 1123 ± 17 and 1141 ± 18 (Chalapathi Rao et al., in press). It should be noted that older $^{40}Ar/^{39}Ar$ age (phlogopite separates) of 1401 ± 5 Ma and 1417 ± 8 Ma were reported for the Kotakonda kimberlite (NKF) and Chelima (Cuddapah basin) lamproite respectively (Chalapathi Rao et al., 1999). The discrepancy between the Rb–Sr and $^{40}Ar/^{39}Ar$ ages from Kotakonda are addressed recently by Gopalan and Anil Kumar (2008) who applied K–Ca dating to samples from the Kotakonda cluster (NKF) and obtained ages of 1068 ± 19 Ma. Gopalan and Anil Kumar (2008) argued that the $^{40}Ar/^{39}Ar$ results of Chalapathi Rao et al. (1999) are affected by excess argon and the NKF kimberlites was also emplaced at ~1100 Ma. However, lamproites of the Cuddapah basin and the Krishna valley support for older ages of emplacement of alkaline potassic rocks within and around the Cuddapah basin (Kumar et al., 2001; Chalapathi Rao et al., 1996, 1999; Osborne et al., 2011). Mineral $^{40}Ar/^{39}Ar$ ages on phlogopite separates from the Majhgawan diamondiferous ultramafic pipe near Panna, Bundelkhand craton, central India yield a mean age of 1073.5 ± 13.7 Ma (Gregory et al., 2006). This new age for the Majhgawan pipe constrains the ages of Upper Vindhyan sedimentation (Bhander–Rewa) to less than ~1075 Ma (Gregory et al., 2006). Recent age determination on the Mahoba mafic dyke, Bundelkhand

craton (Pradhan et al., 2012) gives 1113 ± 7 Ma that is once again similar to that of the Majhgawan pipe.

3. Sample location and description

Thirty-five oriented block samples were collected from ten kimberlite pipes after the orientation was done using magnetic compass by drawing the North direction on the compass on to the block sample and drawing two horizontal lines on the adjacent sides of the block sample using a spirit level. Twenty four oriented block samples were collected from five kimberlite pipes (one from Chigicherla (CG), two from Gollapalle (G) and two from Kalyandurg (KL)) of WKF. Likewise eight oriented block samples were collected from three kimberlite pipes (two from Narayanpet (NK) and one from Kotakonda (KK)) from NKF. As the kimberlites under study occur as dykes of a few metres, the number of samples incorporated in this study is considered to be sufficient and very good representatives of the outcrops. Furthermore, very good directions and poles obtained (below) firmly attest to this. All of them are relatively fresh and have inequigranular texture imparted by two populations of rounded to anhedral macrocrystic and subhedral to euhedral phenocrystic olivine. Olivine macrocrysts/phenocrysts are relatively fresh but some are altered to serpentine. Groundmass phases include serpentine, clinoproxene, phlogopite, carbonate and late-stage phases such as spinel and pervoskite. Rare apatite grains are also present. Detailed petrography, mineral chemistry and whole-rock geochemistry of these kimberlites are reported in Chalapathi Rao et al. (2004), Chalapathi Rao and Srivastava (2009) and Chalapathi Rao and Dongre (2009). Three samples from two kimberlite pipes are also collected from the Siddanpalle (SK) belongs to RKF. The two kimberlite pipes sampled from the RKF (Siddanpalli pipes) for palaeomagnetic studies constitute some of the best preserved samples. These kimberlites consist of olivine, phlogopite and carbonate macrocrysts together with ilmenite embedded in serpentine, perovskite and other opaques. Petrography, whole rock and mineral chemistry of the three pipes near Siddanpalli were reported by Sridhar et al. (2004) and Chalapathi Rao et al. (2010a).

4. Palaeomagnetism and rock magnetism

4.1. Palaeomagnetic experiments

Samples were cored and cut into standard sized cylindrical specimens (25 mm diameter × 22 mm height). A total of 200 specimens were obtained from 35 samples. Natural remanent magnetization (NRM) of all the specimens were measured on a mini spin Molspin and JR-6 Spinner Magnetometers. A minimum of four specimens from each of the sample were treated using stepwise alternating field and progressive thermal demagnetization at a series of applied field/demagnetization steps. Alternating Field demagnetization was conducted using Molspin AF demagnetizer and with applied field up to 150 mT. Thermal demagnetization was conducted up to temperatures of 600 °C with a Magnetic Measurements Thermal demagnetizer (model MMTD – 80). Principal component analysis (Kirschvink, 1980) was used to determine the best-fit lines for each demagnetized sample using Rema 4 software.

4.2. Rock magnetic experiments

The susceptibility of each specimen was measured before treatment to demagnetization using Bartington Susceptibility system (Model MS-2). Isothermal remanent magnetization (IRM) acquisition studies were conducted on each kimberlite pipe using pulse magnetizer (MMPM-9) to further characterize the magnetic behaviour. Curie temperature (K–T) studies were conducted on representative samples using Bartington Susceptibility System (Model MS-2WF and MS-2WFP).

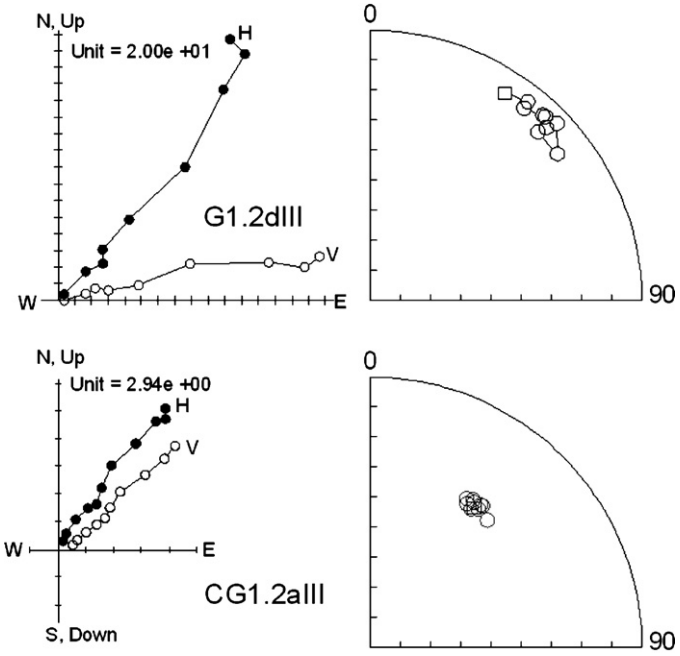


Fig. 2. Orthogonal vector plots from the kimberlites of Eastern Dharwar craton showing typical characteristic remanent magnetization directions. Thermal demagnetization behaviour of kimberlite sample G1.2dIII from Gollapalli and CG1.2aIII from Chigicherla, WKF. Solid circles represent projections on the horizontal plane indicated by 'H'; open circles represent projection onto a vertical plane indicated by 'V'.

5. Results

Out of the thirty five samples which were collected from ten kimberlites, twenty four samples from WKF and eight samples from NKF and three samples from RKF exhibited stable demagnetization behaviour during AFD and thermal treatments. Typical thermal demagnetization behaviour (vector end point diagrams) is shown in Fig. 2. Thermal demagnetization was applied up to temperatures of 600 °C and a loss of 90% of initial NRM strength occurred at about 550–580 °C with a discrete unblocking temperature range between 500 °C and 550 °C as seen in the intensity plots. Alternating demagnetization treatments up to 40 mT removed over 90% of the NRM. Only the samples from WKF give the directions related to 1100 Ma poles whereas rest of the samples (from NKF and RKF) give either steeper inclinations or random directions. The overall mean of characteristic remanent magnetization (ChRM) directions derived using principal component analysis (PCA) gives a mean declination (Dm) = 39°, a mean inclination (Im) = −16° (k = 37.6; α_{95} = 15.2; N = 4). Fig. 3 shows the stereo plot of site mean directions along with overall mean directions with α_{95} for the kimberlite pipes of this study. The mean directions yield a resultant virtual geomagnetic pole (VGP) at 44.5°N and 195.4°E (dp = 8.0/dm = 15.6). Table 1 displays the results of individual sites from this study. The obtained palaeomagnetic directions from this study and Majhgawan ultramafic pipe (central India) as well as other results from reliable data sets of 1100 Ma are listed in Table 2.

Fig. 4 shows the Curie temperature runs (susceptibility vs temperature) and a sharp drop in susceptibility at 565 °C and 595 °C, which is slightly above Curie temperature of magnetite, is observed. Isothermal remanence magnetization (IRM) test curves (Fig. 5) show a smooth rise in intensity and a near saturation

Table 1
Palaeomagnetic results of Eastern Dharwar craton kimberlites of this study.

Location	N/n	Dec	Inc	Kappa (k)	α_{95}	Plat.	Plong.	dp	dm
CG	5/1	32	−23	60.2	9.9	49.2°N	205.8°E	5.6	10.5
G1	6/1	45	−25	33.6	11.7	37.4°N	196.5°E	6.7	12.5
G2	6/1	49	−8	22.4	16.5	38.0°N	184.4°E	8.3	16.6
KL3	4/1	31	−6	–	–	–	–	–	
KL4	3/1	116	−71	95.6	12.6	26.7°S	221.5°E	–	
NK1	3/1	19	−55	842.2	4.2	35.0°N	238.1°E	4.2	6.0
NK9	3/1	55	−68	46.0	18.3	7.4°E	225.6°E	–	
KK6	2/1	90	−65	21.0	17.0	12.2°S	213.5°E	–	
Mean of CG, G1, G2, KL3	21/4	39	−16	37.6	15.2	44.5°N	195.4°E	8.0	15.6

N, number of samples used; *n*, number of kimberlites; Dec, declination; Inc, inclination; *k*, kappa precision parameter; α_{95}, cone of 95% confidence about the mean direction; Plat., pole latitude; Plong., pole longitude.

Table 2
Summary of palaeomagnetic results on Mesoproterozoic kimberlites and other comparable results from India.

Study	N	Dm	Im	k	α_{95}	Pole lat	Pole long	dp	dm
Eastern Dharwar craton									
This study	4	39	−16	37.6	15.2	44.5°N	195.4°E	8.0	15.6
Majhgawan kimberlites									
Gregory et al. (2006)	8	45	−25	37	9.3	33.5°N	203.3°E		
Miller and Hargraves (1994); site 1	6	40	−32	28	13.0	32.4°N	213.2°E		
Miller and Hargraves (1994); site 2	6	27	−22	54	9.2	45.3°N	220.1°E		
Mean of 3 sites	22	38	−26	66	15.3	36.8°N	212.5°E	9.0	16.6
Other comparable results									
Malani mean[a]	4 Studies	359	62	73	8.0	72.7°N	70.5°E	9.5	12.3
Bhander–Rewa mean[b]	18 Sites	34	−20	21	7.0	41.6°N	212.3°E	3.8	7.2
Mahoba dyke[c]	4 Sites	25	−38	36	15.5	38.7°N	229.5°E	9.5	16.3

N, number of sites used; Dm, mean declination; Im, mean inclination; *k*, kappa precision parameter; α_{95}, cone of 95% confidence about the mean direction; dp, semi-axis of the confidence ellipse along the great circle path from site to pole; dm, semi-axis of the confidence ellipse perpendicular to that great-circle path. Malani mean and Bhander–Rewa mean are from Malone et al. (2008).

[a] Mean reported from Klootwijk (1975), Athavale et al. (1963), Torsvik et al. (2001) and Gregory et al. (2005).

[b] Mean reported from McElhinny et al. (1978).

[c] Mean reported from Pradhan et al. (2012).

at 180–200 mT. The rock magnetic tests and demagnetization behaviour show that both the characteristics are of magnetite as the principal remanence carrier with a small contribution from hematite.

6. Discussion

6.1. Kimberlites, Palaeomagnetism and closure age of the 'Purana' sedimentary basins

Kimberlites are widely regarded to carry stable remanent magnetization that was primary and acquired penecontemporaneously along with their emplacement (Hargraves, 1989; Van Fossen and Kent, 1993; Kravchinsky et al., 2002). The first quantitative

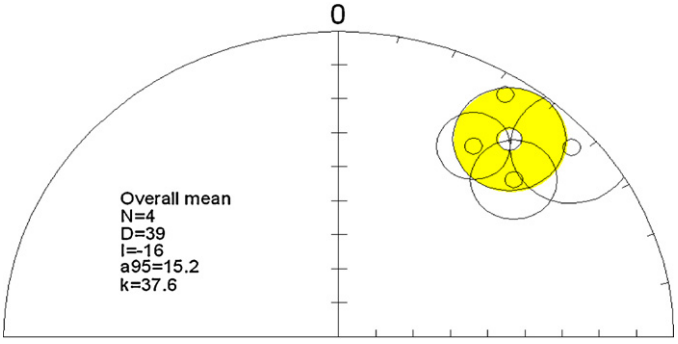


Fig. 3. Site means along with circle of confidence (α_{95}) for four sites and the white circle with yellow colour is the overall mean of four sites with (α_{95}) for the eastern Dharwar kimberlites collected in this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of the article.)

suggestion that some of the South African kimberlites might be of Precambrian age came from a preliminary palaeomagnetic study from the Premier Mine (Jones, 1968) and simultaneous geochronological studies (Allsopp et al., 1967). These studies reveal that individual pipes can give acceptable palaeomagnetic poles as these are distinct from virtual geomagnetic poles (McFadden, 1977; McFadden and Jones, 1977; Hargraves and Onstott, 1980; Kravchinsky et al., 2002). Miller and Hargraves (1994) conducted a palaeomagnetic study of the Majhgawan diatreme of central India and reported a palaeomagnetic pole at 38.9°N, 216.5°E. Palaeomagnetic data for some WKF samples by Miller and Hargraves (1994) yielded a mean declination of 37.5 and a mean inclination of −26.5 (k = 66 and α_{95} = 15.3) and VGP at 36.8°N, 212.5°E (dp = 9.0, dm = 16.6). Pipe-wise mean NRM intensity and susceptibility values vary between 0.70–34.6 × 10^{-3} A/m and 1.8–7.5 × 10^{-6} SI units respectively. Similar NRM intensity and susceptibilities were also reported from South African kimberlites by Hargraves and Onstott (1980) which vary between 1.3–25.3 × 10^{-3} emu/cc and 1.3–3.7 × 10^{-3} G/Oe respectively. Doppelhammer and Hargraves (1994) reported palaeomagnetic data from the Franspoort and Schuller pipes within the Premier kimberlite swarm and combined their results with poles from the National and Premier pipes southern Africa (Jones, 1968; Hargraves and Onstott, 1980). The Wajrakarur and Narayanpet kimberlites of this study exhibit blocking temperatures of 550 °C and 580 °C indicating that magnetite is the primary remanent carrier in these rocks. Interestingly, the available data for South African kimberlites also reveal Curie points between 420 °C and 590 °C (Jones, 1968).

Malone et al. (2008) conducted a palaeomagnetic study on the Bhander–Rewa sedimentary Groups (Upper Vindhyan) and obtained a mean palaeomagnetic pole at 44°S, 34.0°E (α_{95} = 4.3). We note that the VGP obtained for the ~1113 Ma Mahoba dykes

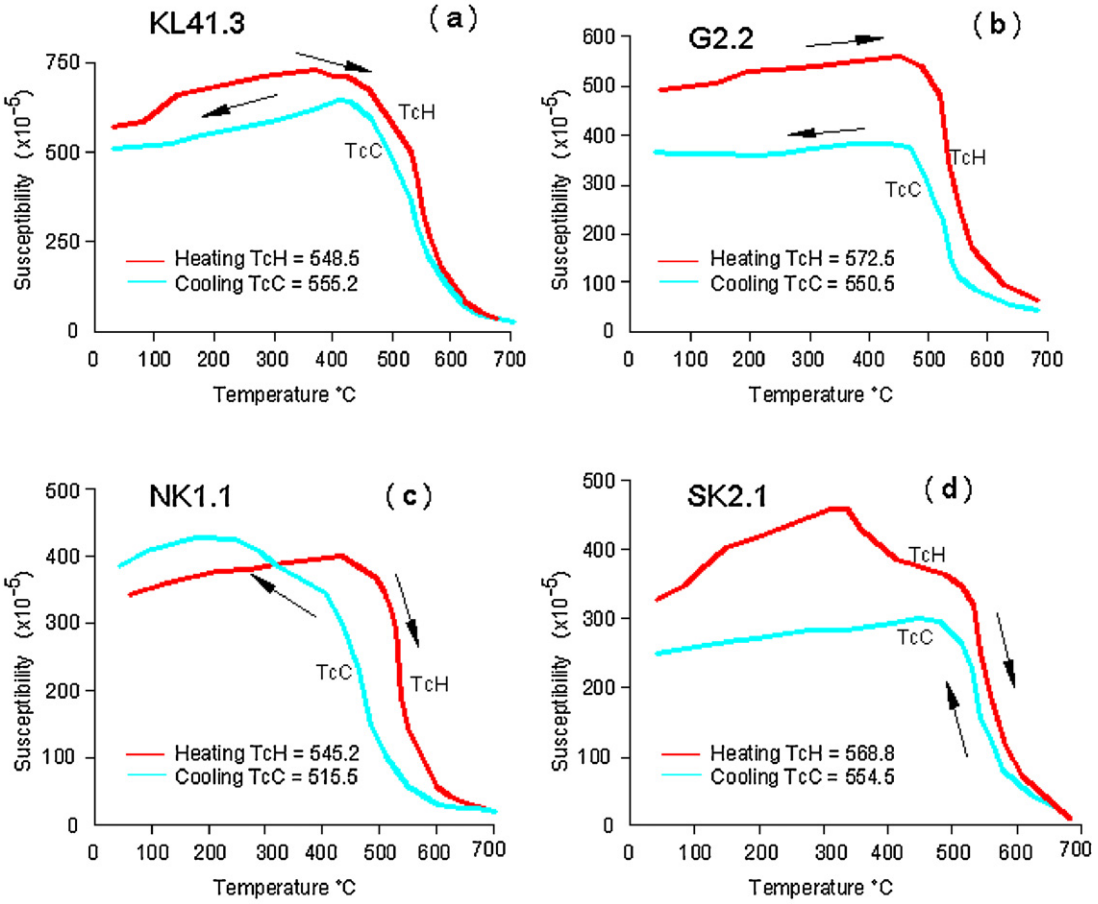


Fig. 4. Curie temperature runs for Anantapur kimberlite samples KL4 1.3 (a) and G2.2 (b), Curie temperature runs for the Narayanpet kimberlite sample NK1.1 (c) and Curie temperature runs for the Siddanpalli kimberlite sample SK2.1 (d).

(37.8°S, 49.5°E) in the Bundelkhand craton generated by Pradhan et al. (2012) and the pene-contemporaneous Majhgawan diatreme are nearly identical to the palaeomagnetic poles obtained from the Bhander–Rewa sedimentary Groups in the Upper Vindhyan sequence (Fig. 6). The VGP (45°N, 195°E) obtained for the ~1100 Ma kimberlites of the present study is also nearly identical to the mean palaeomagnetic pole from the Bhander–Rewa Groups, Mahoba dykes and the Majhgawan diatreme (Gregory et al., 2006; Fig. 6).

The palaeomagnetic results obtained from this study have important relevance to the closure age of 'Purana' Proterozoic sedimentary basins in India viz., the Upper Vindhyan, Chhattisgarh and the Indravati–all of which are considered to be homotaxial based on lithostratigraphy and paleobiological evidences. Originally, the upper most sequences of these basins were thought to be of late Neoproterozoic (Ediacaran) age based on stable isotope systematics, strontium isotope excursions and fossil impressions (see Naqvi, 2005). However, recent studies involving U–Pb dating of zircons in the rhyolitic tuffs from the upper stratigraphic horizons of the Chhattisgarh basin (Patranabis-Deb et al., 2007; Basu et al., 2008; Bickford et al., 2011) and from the Indravati basin (Mukherjee et al., 2012) reveals that the age of youngest deposition to be >1000 Ma. Likewise, from a combination of palaeomagnetic and radiometric dating studies Gregory et al. (2006), Malone et al. (2008) and Pradhan et al. (2012) have opined that the Upper Vindhyan sedimentation closed by at least 1000 Ma. The similarity of the poles obtained from this study with those from Upper Vindhyan sequence, Mahoba dyke and Majhgawan diatreme in Bundelkhand craton supports these older Mesoproterozoic ages and add to their growing data base.

6.2. India in Rodinia supercontinent at 1100 Ma

The late Mesoproterozoic (ca. 1100 Ma) has been postulated as the time interval for the formation of the supercontinent Rodinia (e.g., McMenamin and McMenamin, 1990; Dalziel, 1991; Moores, 1991; Hoffman, 1991; Powell et al., 1993a,b; Torsvik et al., 1996; Weil et al., 1998; Condie, 2001; Rogers and Santosh, 2003, 2004; Meert and Torsvik, 2003; Li et al., 2008; Meert, 2012). The presence of a number of 1300–900 Ma old orogenic/mobile belts (Dewey and Burke, 1973) and correlations between the various cratonic nuclei (Young, 1995; Dalziel, 1997; Karlstrom et al., 1999; Dalziel et al., 2000) support the existence of Rodinia. However, the geometry/paleogeography and duration of the Rodinia supercontinent remains unsettled due to a paucity of well dated, high quality palaeomagnetic poles from various continental components forming the supercontinent (Weil et al., 1998; Meert and Powell, 2001; Meert, 2001; Meert and Torsvik, 2003; Li et al., 2008). Even though, the position of India within the 'Rodinia' assembly is still conjectural (Chakraborty et al., 2010) the similarity between the palaeomagnetic results from northern Bundelkhand (Majhgawan, Mahoba and Bhander–Rewa) and southern eastern Dharwar Indian cratons, as brought out from this study, demonstrate that these two cratons were in reasonable proximity at 1.0 Ga.

In the Indian shield, the ca. 1.0–1.1 orogenic events are also most pronounced in the Eastern Ghats Mobile belt (EGMB), and to some extent across the Central Indian Tectonic Zone (CITZ) and in the Aravalli region in NW India (Mezger and Cosca, 1999; Biswal et al., 2007). Various authors have attached some significance to this 1.0 Ga orogenic event as resulting from the unification of the

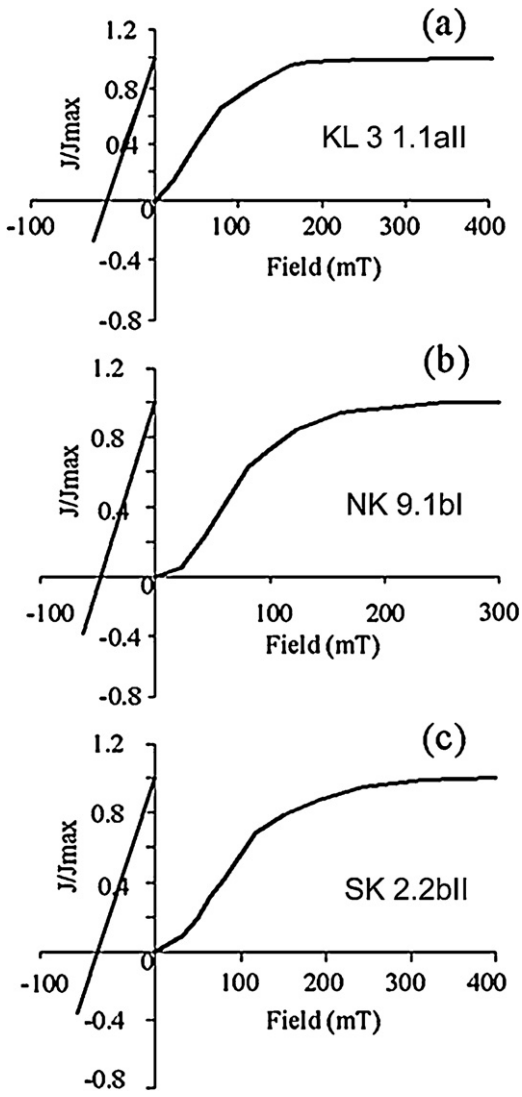


Fig. 5. Isothermal remanence magnetization (IRM) acquisition and back field demagnetization behaviour of (a) Anantapur kimberlite KL3 1.1aII; (b) Narayanpet kimberlite NK 9.1bI; (c) Siddanpalli kimberlite SK 2.2bII. All samples saturate at about 100–200 mT. Coercivity of remanence values ranged from 50 to 60 mT.

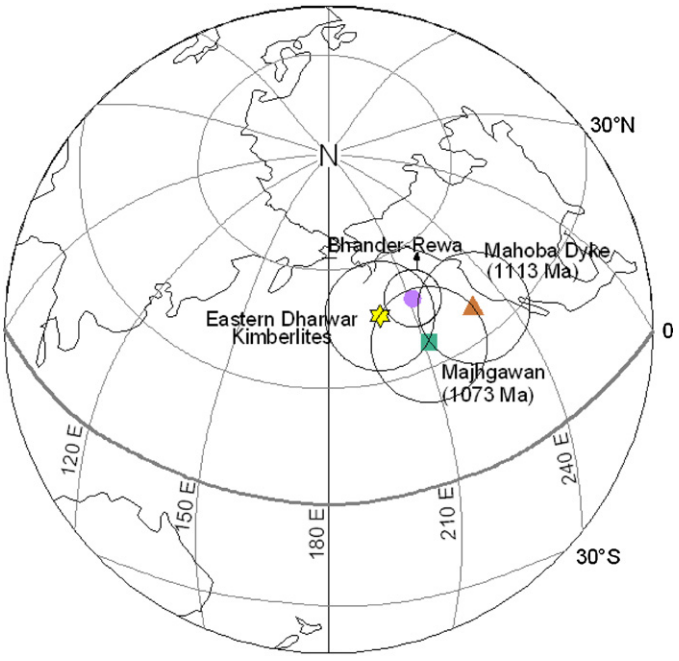


Fig. 6. Palaeomagnetic and virtual geomagnetic poles (VGPs) for the Eastern Dharwar kimberlites (this study), the Majhgawan kimberlite (Gregory et al., 2006), the Bhander–Rewa Groups of the Upper Vindhyan sequence (Malone et al., 2008) and palaeomagnetic pole of Mahoba dykes, Bundelkhand craton (Pradhan et al., 2012).

northern and southern Indian cratons (Singh et al., 2010 and references therein). But the record of such events in the CITZ remains controversial and older accretionary ages of ca. 1.68 Ga are also reported (Bhowmik et al., 2011). However, as the palaeomagnetic poles reported in this study from the Eastern Dharwar craton matches well with those reported from the Bundelkhand craton, we infer that the northern and southern Indian blocks were unified prior to 1.1 Ga. Our finding contrasts with several tectonic models which consider these two were distinct blocks at that time (see Meert et al., 2010 for a review).

The most important outcome of the present study is a firm constrain on the paleoposition of Indian sub-continent in Rodinia configuration at ~1.1 Ga. The recent palaeomagnetic and geochronological studies from the Majhgawan kimberlite (Gregory et al., 2006), Bhander–Rewa Groups of Upper Vindhyan (Malone et al., 2008) and Mahoba dolerite dykes (Pradhan et al., 2012) provided high quality poles to constrain the paleogeographic position of India at 1.1 Ga. A key, well-dated palaeomagnetic pole, from the Kalahari craton at ~1.1 Ga is based on the Umkondo dolerites and the Kalkpunt poles (Hanson et al., 2004; Gose et al., 2004, 2006; Jacobs et al., 2008). The precisely dated Umkondo dolerite palaeomagnetic data from the Kalahari craton are widely correlated with Keweenawan igneous suite of Laurentia (Hanson et al., 2004; Jacobs et al., 2008). These two coeval palaeomagnetic poles are predominantly of one polarity and thus constrain the relative paleolatitudes of the two cratons within the Rodinia configuration. Jones et al. (2003) provided evidence that the ~1.13 Ga Borgmassivet intrusions (Grunehogna crustal province of Antarctica), which are a part of the Umkondo LIP, also carry primary magnetization. Therefore it can be presumed that the Borgmassivet intrusions and the Umkondo mafic rocks in southern Africa acquired their characteristic remanence at the same time. Thus the only reasonably well-dated palaeomagnetic poles from the Kalahari craton that are of direct use in Rodinia reconstructions are the ~1.1 Ga Umkondo and Kalkpunt poles and the ~1.0 Ga poles from the Namaqua–Natal belt. Fig. 7 shows the paleogeographic reconstruction of Rodinia at ~1.1 Ma utilizing the palaeomagnetic poles from the above mentioned cratonic blocks. The configuration, reconstructed for a time-window of 1100 Ma, very likely represents the first phase of Rodinia assembly (see Pesonen et al., 2003). Interestingly, there are a number of palaeomagnetic poles available from other domains of the Rodinia supercontinent that are more or less coeval with those from India (see Pradhan et al., 2012 for a recent compilation).

Studies on Mesozoic kimberlites from southern Africa resulted in a revised apparent polar wandering path (APWP) for Africa in Upper Mesozoic suggesting that kimberlite emplacement does coincide with intervals of rapid plate motion, though this cannot be the primary cause of their intrusion (Hargraves and Onstott, 1980). An examination of the Indian kimberlite palaeomagnetic data listed in Table 2 reveal that there is no mutually agreeable result as reported from such studies. Anand (1971) and Miller and Hargraves (1994) reported SE pointed declinations with upward pointing intermediate and steep inclinations for Wajrakarur pipe 1 and Lattavaram pipe 4 (WKF) respectively. Similar mean directions are also obtained from Kalyandurg (Dm = 116, Im = −71) and Kotakonda (Dm = 90, Im = −65) kimberlites in the present study. Poornachandra Rao et al. (1984) obtained NW pointed declinations with downward inclination for Muligiripalle pipe 5 (WKF). We

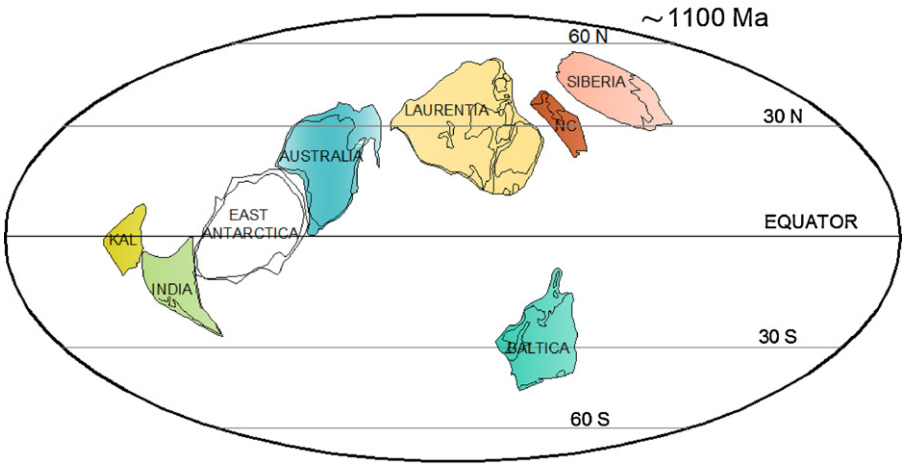


Fig. 7. Palaeomagnetically based reconstruction at 1.1 Ga based on our virtual geomagnetic pole (VGP) from Eastern Dharwar craton (this study); combined virtual geomagnetic pole (VGP) from ~1113 Ma Mahoba dykes of the Bundelkhand craton; Majhgawan kimberlite pole (Gregory et al., 2006) and Bhander–Rewa pole (Malone et al., 2008) and data reported in Pesonen et al. (2003) that places India at low southern latitudinal position.

report similar mean directions (Dm = 321/345, Im = 60/−20) from Siddanpalli (RKF) kimberlite pipe in this study.

Pipe 5 Muligiripalle (WKF) has been dated to be of 1140 ± 14 Ma by Crawford and Compston (1973) and 1153 ± 17 Ma by Chalapathi Rao et al. (1996). Radiometric dating of WKF pipes by Anil Kumar et al. (1993, 2007) showed them to have been emplaced between 1091 ± 20 Ma and 1093 ± 20 Ma suggesting their episodic nature. Radiometric dating of Narayanpet kimberlite pipe by Gopalan and Anil Kumar (2008) as well as newly generated high precision U–Pb (perovskite) data (Chalapathi Rao et al., in press) for several previously undated kimberlites from WKF and NKF indeed gives an age of 1100 Ma thereby highlighting contemporaneous emplacement of WKF and NKF pipes. Chalapathi Rao et al. (1996, 1999) suggested two phases of kimberlite–lamproite emplacements in and around Cuddapah basin separated by a period >300 Ma between 1090 Ma and 1380 Ma. Recent Ar–Ar age of 1400–1500 Ma (Osborne et al., 2011; Phillips, 2012) for Pochampalle lamproite (Krishna lamproites) indeed points out to a much older lamproite emplacement age. More precise geochronological and palaeomagnetic studies on the Cuddapah and Krishna lamproites from the Eastern Dharwar craton are needed to fully evaluate these aspects.

7. Conclusions

This study provides a refined palaeomagnetic pole for the Eastern Dharwar craton kimberlites at 44.5°N, 195.4°E (α_{95} = 15.2°). This palaeomagnetic pole correlates well with the VGP generated from the (i) Majhgawan kimberlite that lies at 36.8°N, 212.5°E (α_{95} = 15.3°) (Gregory et al., 2006), (ii) the Bhander–Rewa Groups in the Upper Vindhyan sequence (Malone et al., 2008) and (iii) Mahoba dyke (Pradhan et al., 2012) – all within the Bundelkhand craton of Northern India. The similarity in palaeomagnetic poles from Eastern Dharwar and Bundelkhand cratons support (i) a Mesoproterozoic closure age of the upper sedimentary horizons for the 'Purana' sedimentary basins and contribute to the growing data base of such ages and (ii) accretion of the northern and southern Indian blocks prior to 1.1 Ga contrary to several earlier tectonic models which consider them to be distinct blocks at that time. Although India's exact position in the Rodinia supercontinent still remains uncertain it is increasingly becoming evident that in ~1.1 Ga palaeomagnetically based reconstructions, India occupies a lower palaeolatitudinal position, much separated from Australia, and East Gondwana very likely did not coalesce until the ~750 Ma (Meert, 2003).

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